

Larger steps in detector optimization

A fourfold increase in Adam learning rates improves feasible loss on two fresh detector topologies in a fixed 600-second diagnostic.

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Result

The four-run diagnostic completed and passed local Objective-checkpoint replay. Fixed 4x learning rates beat the stock rates on both fresh topologies: mean feasible loss was **2.208027348** versus baseline **3.821839915**, with a mean paired difference of **-1.613812567**. This is a useful two-topology mechanism lead. There was no promotion criterion at n=2, and no optimizer was uploaded or assigned a new public score.

The replacement rental was deleted and cleanup verified. Its conservative cost bound is **\$1.37**; an earlier provider failure before container creation adds **\$0.14**. Including the two preceding studies, the combined conservative bound is **\$13.78**, below the \$14 ceiling. These are allocation-time bounds, not settled provider invoices.

Frozen comparison and endpoints

The prepared plan selected the 600-second option solely from the preceding study's verified conservative cost. Both arms use the unchanged BatchedRestartAdam, optimizer seed **91**, population 8, stock anchor plus paired random starts, patience 600 and the same public pre-clock warmup. The sole optimizer-setting change is geometric learning-rate endpoints **0.03–0.15** versus **0.12–0.60**. There is no adaptive gain, polishing, new restart rule or extra evaluation mechanism.

The D/H panel remained untouched through all preparatory revisions and the failed provider start. Generator seeds are 2026091504 and 2026091501; topology identity, initial-array hashes and runtime pair exactly within each block. Known local panels/plans were excluded by exact identity. No official-

archive or hidden-panel exclusion is claimed.

Topology	Baseline	Fixed 4x rates	4x minus baseline
1 (D)	3.541691990	1.367611327	-2.174080663
2 (H)	4.101987839	3.048443369	-1.053544470
Mean	3.821839915	2.208027348	-1.613812567

All **four runs** returned finite feasible results within their 600-second logged budgets, totaling **15,640 evaluations**. The run order was D baseline, D rate_4x, H rate_4x, H baseline. Each worker had fresh process and optimizer state; only compilation was cached. There was no scored retry, substituted topology or source change.

Evaluation matching, prefixes and population activity

Arm	Evaluations	Mean evaluations per logged second	Mean finite-feasible fraction
Baseline	7,792	6.4984	59.73%
Fixed 4x rates	7,848	6.5453	49.63%

The final minima are unchanged when each pair is truncated to its last common evaluation count: **3,904** for topology 1 and **3,888** for topology 2. The observed endpoint lead therefore persists after matching evaluations. The 4x arm produced a smaller feasible-point fraction, showing that more feasible evaluations alone are not the explanation for its better minimum.

Replayed logged prefix	Baseline mean	Fixed 4x mean
150 s	4.309412328	4.255287931
300 s	4.245789772	2.943473531
450 s	3.911687584	2.328010974
600 s	3.821839915	2.208027348

These prefixes reuse the same four histories; they are not extra independent experiments. The panel mean already favored 4x at 150 seconds, but most of the gap appeared later. The 4x final minima first occurred at **592.74 seconds** on topology 1 and **248.15 seconds** on topology 2. Baseline final minima first occurred at 570.77 and 371.51 seconds.

The winning 4x points came from non-anchor lanes **6 and 7**, respectively (zero-based indexing). Each 4x run had incumbent improvements from five different lane positions. This diagnostic exercised population exploration, rather than only the common anchor trajectory seen in some earlier short CPU checks. It does not establish how the same lanes or rates behave on new seeds.

Limits on interpretation

This experiment has two topology units and one paired optimizer seed per unit. It provides neither a calibrated significance claim nor an estimate of within-topology seed variability. The settings are deliberately simple, but the physics trajectories remain sensitive to numerical and stochastic variation. Relevant longer-budget comparisons and fresh topologies/seeds are still needed before promotion.

Do not rank fixed 4x rates directly against sign-adaptive Adam from the preceding study. Those results used different topologies, optimizer seeds and a different GPU driver. Their absolute means are not a paired comparison. This diagnostic also cannot determine whether constant larger steps explain the adaptive arm's earlier gain; that requires comparing them on a common fresh panel with matched budgets.

The result is at **600 seconds**, not the competition's four-hour budget. It is not a public-leaderboard score, an estimate of hidden-score improvement, or authorization to submit or spend further. The retained package remains unchanged.

Provider failure, replacement and cleanup

The first host repeatedly reported a missing Docker layer while trying to create its container. It never created a usable container, started the experiment or logged a physics evaluation. Its initialization log and failed-start-cost.json are preserved under artifacts/generated/rate-uifo-20260905-v3-600/rental/. Cleanup was verified at **2026-09-05 03:28:53.9914814 UTC**. The 308.678137-second conservative allocation interval contributes the separately recorded **\$0.14** bound.

The coordinator copied the same frozen plan and source bundle to a replacement host. Their bytes and source-member hashes match the previously prepared artifacts. This was a replacement of failed infrastructure before physics, not a retry of an unfavorable scored run. All four scored runs then completed once, and cleanup was verified at **2026-09-05 04:21:26.265421 UTC**.

The replacement interval from create intent through verified deletion was **3,069.808788 seconds**, within its 55-minute ceiling. At the conservative all-in hourly bound of \$1.60, the rounded bound is **\$1.37**. Worker wall time totaled **2,818.06 seconds**; per-run overhead outside logged search ranged from **52.10 to 164.63 seconds**. The failed first allocation is charged separately rather than hidden inside the replacement deadline.

The final combined bounds are:

Allocation	Conservative bound
Prior H100 study	\$8.51
Continuous-explorer A100 study	\$3.76
Failed rate-study provider start	\$0.14
Completed replacement rate study	\$1.37
Total	\$13.78

The coordinator verified an empty pod inventory and zero current compute spend per hour after cleanup. Both local state receipts record `cleanup_verified=true`. Provider-ledger reconciliation remains separate from these conservative allocation bounds and does not require keeping hardware running.

Runtime and evidence artifacts

The four workers reported **NVIDIA A100-SXM4-80GB**, 81,920 MiB, MIG Disabled, driver **580.159.04**, Python 3.12.13, JAX 0.9.0.1, dfbench 0.3.3 and x64 enabled. The independent history auditor verifies the saved Objective NPZ losses, feasibility, counts, timestamps, initial parameters, source/runtime pairing and minima against JSON. In particular, unconstrained checkpoint `best_loss` is never substituted for the finite-feasible score. The coordinator’s local replay passed all four runs and reproduced the GPU report.

- Plan SHA-256: c5c315c78c78165a57e916fcc839e637a15600b33e50cc11dd4f97fb7ece1e90.

- Frozen source-bundle SHA-256:
331c3f2ea48d4a35e099a47e2b744cb310637a738e465433df8f1c3e2d6511db.
- Local audited analysis SHA-256:
2acb653e12aabc8285b793a9db078c7d85265eb14390ecbf7f672b3a5aa6c5b.

Raw and derived results are under `artifacts/generated/rate-uifo-20260905-v3-600-host2/results/`:

- `analysis.json`: four-run NPZ replay, paired differences and explicit no-promotion/no-quality-rejection status.
- `final-comparison.csv`: endpoint, evaluation count, timing, feasible fraction, common-evaluation minimum and incumbent-lane details.
- `budget-ladder.json`: all four time prefixes and matched-evaluation results.
- `progress-time.png` and `progress-evaluations.png`: complete two-topology plots.
- `final-data-summary.json` and `derived-artifacts-manifest.json`: aggregate data, verification results and derived-file hashes.

The existing plotting and prefix tools generated these artifacts. All four plotted minima match the audited endpoints, and both rendered figures were visually inspected. Generated histories and figures remain outside Git. This completed comparison changed no frozen optimizer source or uploaded package.

The 47-member evidence archive is 24,176,132 bytes, SHA-256
9a5bafb65bac9687f7557c64172fc47c5319b617cce45344064c8571aa43f98. ZIP integrity, every member hash, and an identical second local copy under `learn2design-runpod-results/rate-uifo-20260905/` were verified. The archive includes the provider-startup failure receipts separately from the four completed scientific runs.

Publication record

This edition was selected for Lyrebird on September 22, 2026. The source manuscript is preserved with its original scope. Relative evidence links resolve to the source repository.

Research assistance: Codex. Alireza Afshan is the project owner and named author; assistance is disclosed and does not constitute independent review. No journal acceptance or DOI is claimed.

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